20/02/2025

INTRODUCTION

DAVID Mihnea-Stefan
AnW - 2025

ABOUT ME

INFORMATIONS

- Name: DAVID Mihnea-Stefan
- Fourth semester - BSc CS at ETH Zurich
- Soon : University of Pennsylvania (UPenn)
- Soon: FernHagen - Political and Educational Sciences
- Background & Education

HOBBIES & INTERESTS

- Math & Algorithms & Theoretical CS & Physics
- Psychology, Sociology, and Politics
- Educational Sciences in the Pre- and University

Contact Informations

- Email: mihdavid@ethz.ch
- WhatsApp : next Slide
- Instagram: @david_mihnea

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WhatsApp Group : AnW-G15-2025

Why do we need it?

- Informal communication environment
- Questions and answers (both from me and from you)
- A good way to get to know each other better
- Faster communication
- Homework 😈

Organization

- Every week from 16:15 to 18:00
- LEE D 105 (English)

Main Topics

1	**Continuation of AnD - Graph Properties + P vs. NP in Practice**
2	**Matching Algorithms + Coloring Algorithms + Their Role in Problem Solving**
3	**Introduction to Probabilities + Types of Distributions + Classic Problems**
4	**Randomized Algorithms; Geometric Algorithms**
5	**Flow Algorithms and Flow Networks**

*** I will focus on these topics in a way that highlights the most important aspects for the exam while also covering the topics that you find more challenging or that require extra emphasis.

*** Don't hesitate to come up with suggestions

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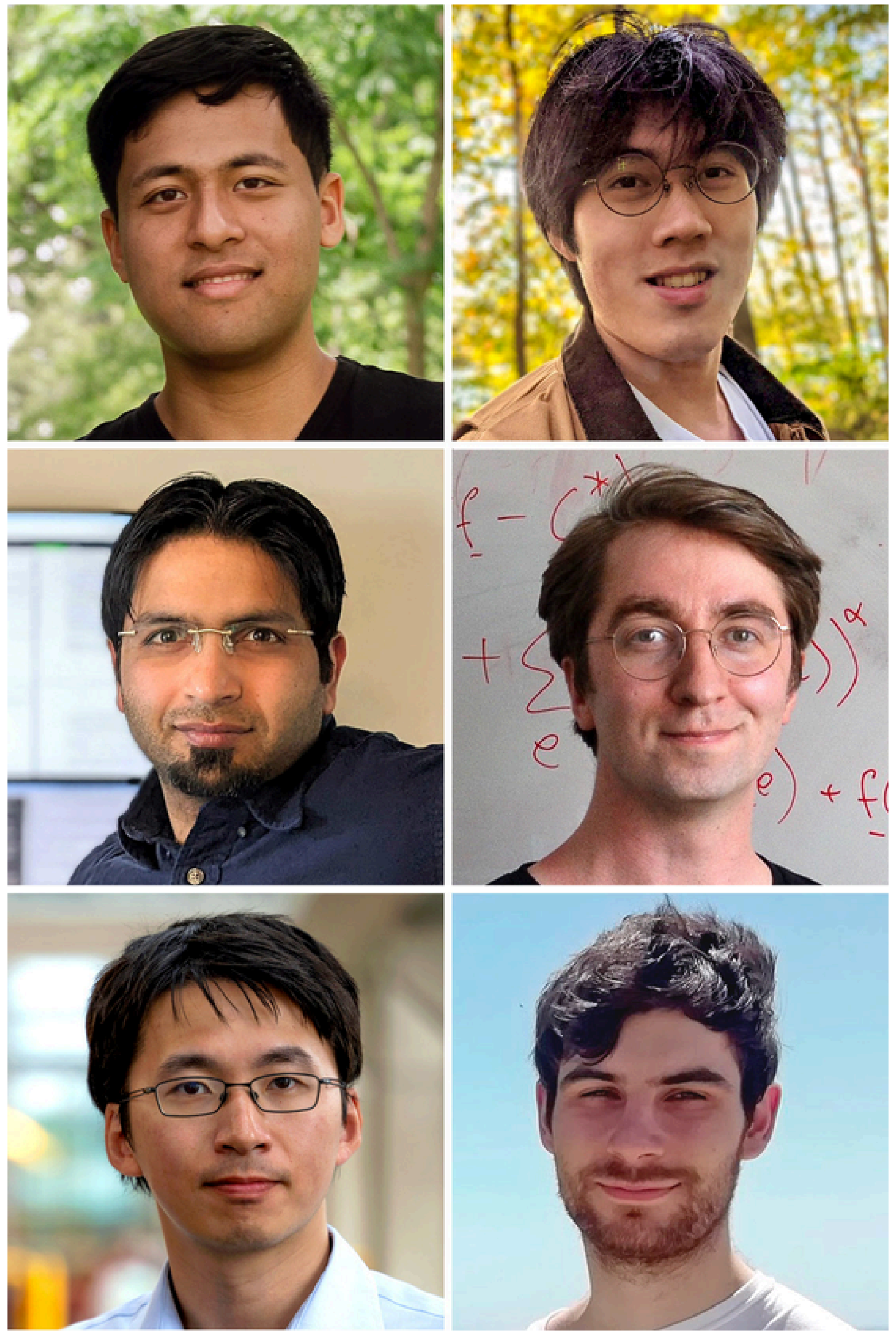
Why are flow algorithms so important in this course?

Well….



NETWORKS

Researchers Achieve 'Absurdly Fast' Algorithm for Network Flow

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Computer scientists can now solve a decades-old problem in practically the time it takes to write it down.

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Some thoughts on the pressure imposed by the first year

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Structure of the course

1. **Script and lecture slides + quizzes**

2. **Programming Homework**

   From the second week on (i.e. starting on February 27th), the students will be assigned a programming task on CodeExpert.

   For solving each of those programming exercises they can get up to two bonus points. Those will be graded automatically.

3. **Theoretical Homework**

   From the second week on (starting on February 27th), every two weeks (even weeks) there will be a theoretical homework to solve. The students have a week to solve those (until 10:00 on the following Thursday). The students should submit solutions via Moodle, and it is your responsibility as a TA to grade them.

   For each theoretical homework, the students can get up to two bonus points

Structure of the course

4. **Peer graded theoretical homeworks**

Every two weeks, starting on the third week of classes (odd weeks), there will be a peer graded theoretical homework. The students should submit the solutions via Moodle, the Moodle will shuffle them to appropriate students for grading, and then students should upload - again on Moodle - the graded homework

The students can get up to two bonus points

5. **Moodle mini-quizes**

Starting on the second week of the classes (February 27th), every even week, there will be a short Moodle quiz, that students will solve during the first ~5 minutes of the exercise class.

The students can get up to two bonus points for solving this quiz.

**AnD Style

Bonus points

Collecting bonus can raise the students grade by at most 0.25 points

i.e. final grade will be min(6, exam grade + 0.25 * ***alfa***), where

alfa := min(1, students bonus points / (0.8 * max possible bonus points)

** ***Almost everyone will get the full bonus… so it is not an advantage to have it, but it can be a disadvantage not to have the 0.25 bonus***

My experience with this course...

Questions?